「Thm. Let $f$ be a Holomorphic on $\overline{B_r(z_0)}$. Then, for any $n \in \mathbb{N}$,

$$|f^{(n)}(z_0)| \le \frac{n!}{r^n} \cdot \sup_C |f| \quad \text{where } C \text{ is a curve along } \partial B_r(z_0).$$

Prof. By the generalized Cauchy's formula,

$$|f^{(n)}(z_0)| = \left| \frac{n!}{2\pi i} \cdot \int_C \frac{f(z)}{(z-z_0)^{n+1}} dz \right|$$

$$\le \frac{n!}{2\pi} \cdot \int_C \frac{|f(z)|}{|z-z_0|^{n+1}} |dz|$$

$$\le \frac{n!}{2\pi} \cdot \frac{M}{r^{n+1}} \cdot 2\pi r = \frac{n! M}{r^n}. \quad (M = \sup_C |f|)$$

「Corollary. Let $f$ be an holomorphic on $B_r(z_0)$.
If $|f(z)-b| \le M$, then $|f'(z_0)| \le \frac{M}{r}$.
Prof. Let $g(z) = f(z) - b$. Then clearly $g$ is analytic and given $0 < \delta < r$,

$$|g(z)| \le M \text{ for all } z \in B_\delta(z_0).$$

Therefore, $|f'(z_0)| = |g'(z_0)| \le \frac{M}{\delta} \to \frac{M}{r}$ as $\delta \to r$.

「Liouville's Theorem.
A bounded entire function must be a constant.
prof. $|f'(z_0)| \le \frac{1}{r} \cdot \sup|f| \to 0$ as $r \to \infty$, for any $z_0 \in \mathbb{C}$. So $f' \equiv 0$, $f$ is constant

「Generalized Liouville's Theorem.
Let $f$ be a non-constant entire function. Then, for any $a \in \mathbb{C}$,
Given $\varepsilon > 0$, there is $z_0 \in \mathbb{C}$ such that $|f(z_0) - a| < \varepsilon$.

Proof. If there is $a \in \mathbb{C}$ such that $|f(z) - a| \ge \varepsilon$ for all $z$,
then $g(z) = \dfrac{1}{f(z)-a}$ is entire, bounded by $\frac{1}{\varepsilon}$.
So $g$ must be constant, so is $f$. It is contradiction.

Thm. Let $f$ be an entire function satisfying: for some $\lambda > 0$, and for some $r_0 \geq 0$,

$$|f(z)| \leq M r^{\lambda} \text{ for all } |z| = r \geq r_0.$$

Then, $f$ is a polynomial such that $\deg f \leq \lambda$.

Proof. Since $f$ is entire, for all $z \in \mathbb{C}$, the taylor expansion

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \cdot z^n$$

is valid. By the Cauchy's inequality, for each $n \in \mathbb{N}$,

$$\frac{|f^{(n)}(0)|}{n!} \leq \frac{1}{n!} \cdot \frac{M r^{\lambda} \cdot n!}{r^n} = \frac{M}{r^{n-\lambda}} \to 0 \text{ as } r \to \infty, \text{ if } n - \lambda > 0.$$

So, $f(z)$ is a polynomial of degree at most $\lambda$. ▨

Prop. Let $f$ be an entire such that for some $a \geq 0$, $b > 0$,

$$|f(z)| \leq a + b \cdot |z|.$$

Then $f$ is either constant or a first degree polynomial.

Proof. Given $z_0 \in \mathbb{C}$, for all $|z - z_0| = R$,

$$|f(z)| \leq a + b|z| = a + b|z - z_0 + z_0| \leq a + b(R + |z_0|)$$

Take $M = a + b(R + |z_0|)$. Then, by the Cauchy inequality

$$\left| \frac{f''(z_0)}{2!} \right| < \frac{2!}{R^2} (a + b(R + |z_0|)) \to 0 \text{ as } R \to \infty.$$

So that $f'' \equiv 0$, since $z_0$ was arbitrary, so $f'$ is constant. ▨

Prop. Let $f$ be an entire such that $|f(z)| \leq |e^z|$.
Then $f(z) = ke^z$ for some $|k| \leq 1$

Proof. Define $g(z) = f(z)/e^z$. Since $f$ is entire and $e^z$ cannot be zero, $g$ is entire. By the hypothesis, $|g(z)| = |f(z)|/|e^z| \leq 1$, so $g$ is a constant the absolute value less than 1, by the Liouville's Theorem. So,

$$g(z) = \frac{f(z)}{e^z} = k \text{ for some } |k| \leq 1, \text{ or } f(z) = ke^z.$$

▨

「Prop.

Let $f$ be an entire function satisfying: for fixed $k \in \mathbb{N}$,

$$|f^{(k)}(z)| \leq |z| \text{ for all } z \in \mathbb{C}.$$

Then,

Proof. Given $z_0 \in \mathbb{C}$ and $R > 0$,

$$|z| = |z - z_0 + z_0| \leq R + |z_0| \text{ for all } |z - z_0| = R.$$

By the generalized Cauchy's inequality,

$$|f^{(n+k)}(z_0)| \leq \frac{n!}{R^n} \cdot \sup_{|z-z_0|=R} |f^{(k)}(z)| \leq \frac{n!}{R^n} \cdot (R + |z_0|).$$

So, for $n \geq 2$, $|f^{(n+k)}(z_0)| = 0$. And, for $n = 1$,

$$|f^{(k+1)}(z_0)| \leq \frac{1}{R} \cdot (R + |z_0|) \to 1 \quad \text{as } R \to \infty.$$

And, by the hypothesis, $|f^{(k)}(0)| \leq 0$. So, substituting $z_0 = 0$, then

$$f(z) = a_{k+1} z^{k+1} + 0 \cdot z^k + a_{k-1} z^{k-1} + \cdots + a_1 z + a_0 \text{ with } |a_{k+1}| \leq \frac{1}{(k+1)!},$$

the reduced polynomial of degree $k+1$.

「Growth Lemma.

Let $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$ with $a_n \neq 0$.
Then there is $r > 1$ such that

$$\frac{|a_n|}{2} \cdot |z|^n \leq |P(z)| \leq \frac{3|a_n|}{2} \cdot |z|^n \quad \text{for all } |z| \geq r.$$

Proof. For $z \neq 0$, we can rewrite

$$P(z) = z^n \cdot \left(a_n + \frac{a_{n-1}}{z} + \frac{a_{n-2}}{z^2} + \cdots + \frac{a_0}{z^n}\right)$$

By the triangle inequality,

$$|z|^n \cdot \left[|a_n| - \left|\frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n}\right|\right] \leq |P(z)| \leq |z|^n \cdot \left[|a_n| + \left|\frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n}\right|\right]$$

For $|z| > 1$,

$$\left|\frac{a_{n-1}}{z} + \frac{a_{n-2}}{z^2} + \cdots + \frac{a_0}{z^n}\right| \leq \frac{1}{|z|} \cdot \left[|a_{n-1}| + \cdots + |a_0|\right]$$

Take $K = |a_{n-1}| + \cdots + |a_0|$. Now, for $|z| \geq \max\left\{1, \frac{2K}{|a_n|}\right\}$, so that $\frac{K}{|z|} \leq \frac{|a_n|}{2}$, the inequality holds. 証

「Corollary. Every non-constant polynomial has at least one zero.